
Decodability Is Not Localization: Auditing Hidden-State Error Probes in Qwen2.5-Math-1.5B-Instruct

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Abstract

1 We audit whether hidden-state probes identify when mathematical reasoning first
2 becomes invalid, or instead exploit easier correlates of an invalid prefix. In a
3 case study of Qwen2.5-Math-1.5B-Instruct on ProcessBench, a linear probe pre-
4 dicts the persistent target *invalid so far* with held-out AUROC 0.866. This num-
5 ber overstates what the representation result establishes: prefix TF-IDF reaches
6 0.760, structural metadata reaches 0.780, and metadata augmented with final-
7 answer correctness reaches 0.857; the last AUROC is not distinguishable from the
8 hidden-state result under a paired whole-trace bootstrap (difference 0.009, 95% CI
9 $[-0.014, 0.032]$). Hidden states retain a larger advantage on trace-level localiza-
10 tion (Process F1 0.393 versus 0.288 for metadata plus outcome; paired difference
11 0.105 $[0.047, 0.163]$), but localize the exact annotated boundary on only 28.9%
12 of erroneous traces and falsely alarm on 38.8% of correct traces. A post-hoc,
13 metadata-matched transition probe obtains AUROC 0.769 on held-out pairs, sug-
14 gesting that an onset-related state change is decodable but not providing confirma-
15 tory evidence. Two verdict readouts fail their behavioral validity checks, so the
16 intervention remains a failed pilot. The supported conclusion is a model-specific
17 gap among prefix decodability, onset localization, calibration, and demonstrated
18 causal use.

19 1 Introduction

20 Process supervision asks whether each step of a solution is valid, rather than whether the final
21 answer happens to be correct (Cobbe et al., 2021; Lightman et al., 2023). A related mechanistic
22 question is whether a solving model’s own representation changes when its reasoning first goes
23 wrong. Probe accuracy alone cannot answer that question. If every boundary after the first error
24 is labeled positive, a classifier may recognize accumulated inconsistency, solution length, generator
25 style, or final-answer outcome without responding at the onset.

26 We separate four claims that are often collapsed: (i) invalid prefixes are linearly decodable; (ii) a
27 score localizes the first error; (iii) the signal transfers across benchmark sources; and (iv) the model
28 behaviorally uses the decoded direction. We test these claims in one explicitly bounded setting:
29 Qwen2.5-Math-1.5B-Instruct on ProcessBench (Yang et al., 2024; Zheng et al., 2025). The primary
30 pipeline uses problem-grouped splits and a persistent invalid-so-far target. We then audit it with
31 prefix-text and metadata controls, equal-trace weighting, trace-macro metrics, source and generator
32 stratification, an onset-aligned analysis, and a matched transition probe.

33 The audit changes the interpretation. The hidden-state probe’s pooled AUROC is high, but a diag-
34 nostic model that is given structural metadata and final-answer correctness nearly matches it. The

Table 1: Closest hidden-state verifier work. “Use” records the reported downstream or causal test, not a judgment of overall paper quality.

Work	Supervision / task	Scope	Onset evaluation	Use test
Sun et al.	arithmetic error	addition, GSM8K	no first-error metric	re-prompting
ReProbe	step credibility	math, planning, QA	step scoring	test-time scaling
STEP	step quality	reasoning traces	early step scoring	pruning
Ours	human first error	four PB sources	first crossing + transition	failed pilot

Table 2: Data flow. Boundary counts are written-step boundaries.

Split	Attempted	Retained	Excluded	Boundaries	Error traces	Correct traces
Train	2,041	2,014	27	14,944	1,311	703
Validation	681	677	4	4,980	444	233
Test	678	669	9	4,985	432	237
All	3,400	3,360	40	24,909	2,187	1,173

hidden state is more useful for the thresholded trace-level task, and its score changes near the human annotation, but localization remains unreliable. The causal assay is not interpretable because its baseline verdict scores rank valid and invalid prefixes below chance. We therefore make one narrow contribution: a worked example showing that strong prefix decoding and cross-source ranking can coexist with weak onset localization, shortcut sensitivity, poor calibration, and no validated causal evidence.

2 Related work and contribution boundary

External process reward models learn step quality from human or model supervision (Lightman et al., 2023; Li et al., 2023). ProcessBench annotates the earliest error in 3,400 model-generated traces from GSM8K, MATH, OlympiadBench, and Omni-MATH (Zheng et al., 2025; Hendrycks et al., 2021; He et al., 2024; Gao et al., 2024). We repartition these benchmark sources for representation analysis; they are heterogeneous sources, not independent mathematical domains or evidence of model-family generalization.

Closest to this work, Sun et al. probe arithmetic errors and use the probe for selective re-prompting; ReProbe learns hidden-state step scorers for test-time scaling; and STEP uses hidden-state scores to prune reasoning traces (Sun et al., 2025; Ni et al., 2026; Liang et al., 2026). Table 1 states the comparison at the level needed for our claim. Our novelty is not that hidden states predict reasoning quality. It is the joint audit of a human first-error target using first-crossing, matched-transition, source-transfer, shortcut, and failed-assay evidence.

Probing work cautions that decodability need not imply behavioral use (Hewitt and Liang, 2019; Belinkov, 2022; Elazar et al., 2021). This distinction is especially important here: an additive intervention is evidence only if its behavioral readout first distinguishes the relevant baseline states.

3 Experimental design

3.1 Data flow, model, and split

Each ProcessBench trace supplies a problem, generator identity, written steps, source, final-answer correctness, and a human first-error index e_i , with $e_i = -1$ for fully correct traces. Traces sharing normalized problem text form a group. Seed 42 assigns groups within source $\times$ error-status strata to 60% train, 20% validation, and 20% test. A 2,048-token limit excludes a complete trace rather than truncating it. Table 2 gives the full split accounting; Appendix A gives source-level counts.

We run Qwen2.5-Math-1.5B-Instruct in bfloat16 and record its 1,536-dimensional residual-stream state at each written step and each of 29 hidden-state indices. The completed extraction used batch size 16 on one NVIDIA A100 GPU. Python 3.10+, PyTorch 2.2–2.x, Transformers 4.45–4.x, and

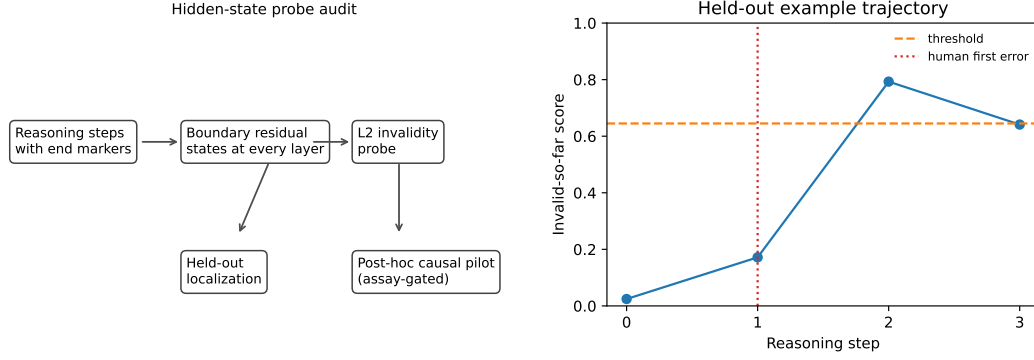


Figure 1: The persistent target and first-crossing evaluation. A causal forward pass produces one boundary state per step. In the example, the score crosses one step after the annotated error.

scikit-learn 1.4–1.x are specified in the accompanying environment. Artifact identity fixes the model, context length, dtype, and dataset SHA-256 (Appendix C).

3.2 Persistent target, probe, and trace-level estimands

At boundary k of trace i , the frozen primary target is

$$y_{ik} = \mathbb{K}[e_i \geq 0 \wedge k \geq e_i]. \quad (1)$$

It means *invalid so far*; it does not label only the first invalid step. A class-balanced, L2 logistic probe is fit separately at each index after train-set standardization. Validation AUROC selects $C \in \{0.01, 0.1, 1, 10\}$; validation Process F1 selects a threshold on a 0.005 grid from 0.05 to 0.95 and then the layer. The selected index is 23, $C = 0.01$, threshold 0.645; it is refit on train plus validation and evaluated once on test.

The predicted first error is the first threshold crossing, or -1 if none occurs. We report boundary AUROC and average precision, exact error-boundary accuracy among erroneous traces, correct-trace rejection, their harmonic mean (Process F1), complete-trace accuracy, and detection within one or two steps. These localization metrics weight traces equally. Primary intervals use 1,000 whole-trace bootstrap draws; the post-hoc paired comparisons use 2,000 draws because they were added in a later audit configuration. We state this difference rather than presenting the two draw counts as one frozen protocol.

3.3 Shortcut and weighting audits

Every added control uses the same train/validation/test groups and selects its threshold on validation only. Current-step TF-IDF uses the destination step. Prefix TF-IDF uses the problem and all reasoning text available through the current boundary. Structural metadata uses source, generator, absolute and fractional step position, trace length, and token count. A deliberately strong diagnostic adds final-answer correctness; because that field can be unavailable online, it is a shortcut audit rather than a deployable baseline. A shuffled-label hidden probe checks the fitting procedure. We report paired probe-minus-control intervals from trace resampling.

The primary objective and boundary AUROC weight each boundary equally. To expose its consequence, we recompute AUROC, average precision, and step F1 using weights inverse to the number of boundaries in each trace. We also stratify trace-level outcomes by source, generator (at least 20 test traces), final-answer correctness, error position, trace-length quartile, and token-count quartile. These subgroups are descriptive, not multiple confirmatory tests.

3.4 Post-hoc onset analyses

The following analyses were designed after the primary test result and are labeled post-hoc. We circularly shift each frozen score trajectory 5,000 times, preserving its values while breaking alignment to the annotation. We compare the score jump at each noninitial error with a transition from a fully

Table 3: Held-out probe and shortcut controls. Paired intervals are hidden minus control.

Predictor	AUROC	Process F1	Δ AUROC [95% CI]	Δ F1 [95% CI]
Hidden state	.866	.393	—	—
Prefix TF-IDF	.760	.281	.106 [.083,.132]	.112 [.058,.166]
Structural metadata	.780	.147	.086 [.065,.107]	.246 [.198,.298]
Metadata + final outcome	.857	.288	.009 [−.014,.032]	.105 [.047,.163]
Current-step TF-IDF	.733	.254	.133 [.111,.155]	.139 [.090,.188]
Position only	.730	.054	.136 [.113,.159]	.339 [.292,.385]
Shuffled hidden labels	.518	.179	.348 [.317,.375]	.214 [.159,.266]

correct trace, preferring the same source and generator and then minimizing differences in relative position, trace length, and token count.

The transition probe is a separate analysis of $h_{i,e_i} - h_{i,e_i-1}$. Every eligible erroneous transition is paired with a correct-trace transition in the same frozen partition. The resulting balanced sets contain 1,152 train, 387 validation, and 380 test pairs. Layer and C are selected by validation AUROC; index 21 and $C = 0.01$ are selected. The test report includes AUROC, average precision, within-pair accuracy, and a two-way bootstrap over error IDs and reused correct-trace IDs. Controls predict from position, destination-step TF-IDF, or shuffled labels. This protocol is now fully specified and executable, but the test set was not untouched when the analysis was conceived.

4 Results

4.1 High prefix AUROC is not specific to hidden-state invalidity

The selected probe reaches AUROC 0.866 (95% CI [0.849, 0.884]), average precision 0.843 ([0.813, 0.870]), and step F1 0.743 ([0.716, 0.769]). Within erroneous traces, first-step centering leaves pooled AUROC at 0.881 ([0.862, 0.900]), and mean per-trace AUROC is 0.968 ([0.958, 0.976]). Stable trace offsets therefore do not explain all discrimination.

The shortcut audit nevertheless makes the pooled result much less specific. Table 3 shows that structural metadata reaches AUROC 0.780, prefix text reaches 0.760, and metadata plus final outcome reaches 0.857. The hidden-minus-outcome-metadata AUROC difference is only 0.009 ([−0.014, 0.032]), while the Process F1 difference is 0.105 ([0.047, 0.163]). Thus hidden states add trace-level localization value under the selected threshold, but pooled ranking alone cannot be interpreted as an isolated representation of mathematical invalidity.

Equal trace weighting changes AUROC by +0.001 (95% paired CI [−0.007, 0.010]), average precision by −0.005 ([−0.018, 0.009]), and step F1 by −0.009 ([−0.021, 0.005]). Boundary weighting is therefore not driving the aggregate result, although length remains an important calibration nuisance. Complete-trace accuracy falls from 0.491 in the shortest length quartile to 0.264 in the longest; correct rejection falls from 0.719 to 0.350. Source-level complete accuracy ranges from 0.298 to 0.512, and generator-level estimates are correspondingly heterogeneous (Appendix B).

4.2 First-error localization remains weak

The hidden score localizes 28.9% of erroneous traces exactly, 59.3% within one step, and 72.7% within two. It crosses somewhere on 88.7% of erroneous traces, but crosses on 38.8% of fully correct traces. Among detections, the signed error averages +0.59 steps ([+0.36, +0.80]), indicating late detection. Process F1 is 0.393 ([0.351, 0.436]) and complete-trace accuracy is 0.404.

Post-hoc temporal tests show more alignment than these threshold metrics alone suggest. Exact localization is 0.289 versus 0.164 under circular shifts ($p = 0.0002$). Across 380 noninitial errors, the mean onset jump is 0.257; matched correct transitions rise 0.113, for a paired difference of 0.144 ([0.096, 0.193]). The mean score continues from 0.341 immediately before the error to 0.559 at onset, 0.680 one step later, and 0.731 two steps later. This is a gradual transition into a persistent state, not a sharp alarm.

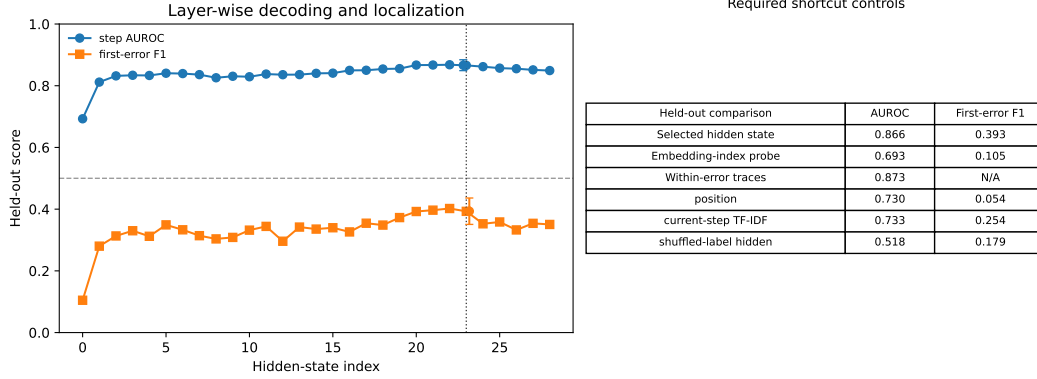


Figure 2: Layer-wise held-out performance and original controls. The embedding-index row is a linear probe at hidden-state index 0, included as a depth comparison rather than a lexical control. Process F1 is N/A within error-only traces because correct-trace rejection is undefined. The stronger prefix and metadata controls appear in Table 3.

138 The matched transition probe reaches AUROC 0.769 ([0.713, 0.820]), average precision 0.741
 139 ([0.672, 0.816]), and paired accuracy 0.753 ([0.676, 0.824]). Position, destination-step TF-IDF, and
 140 shuffled-label controls reach AUROC 0.63, 0.67, and 0.58. This result supports linear decodability
 141 of an onset-related change on the already-inspected split. Replication on an untouched set or second
 142 model is required before treating it as confirmatory.

143 4.3 Source ranking transfers; model generalization is untested

144 Separate probes trained on one ProcessBench source and tested on another have mean off-diagonal
 145 AUROC 0.805 versus 0.845 on the diagonal; every off-diagonal cell exceeds 0.739. Thresholded
 146 Process F1 is less stable: means are 0.261 off diagonal and 0.336 on diagonal, with individual off-
 147 diagonal values from 0.055 to 0.371. These cells show cross-source ranking within one benchmark
 148 construction. They do not establish transfer to another model size, family, or dataset.

149 4.4 The causal experiment is a failed pilot

150 The original CORRECT/INCORRECT readout has baseline AUROC 0.283 and specificity zero. A re-
 151 vised question with conditional Yes/No probabilities remains below chance (AUROC 0.342, speci-
 152 ficity 0.203 on 256 balanced boundaries). The intervention code now requires both AUROC > 0.5
 153 and nonzero specificity before a learned-direction effect is interpreted. Neither readout passes. Ran-
 154 dom orthogonal directions cannot substitute for a positive control of assay sensitivity, so the dose
 155 response is not causal evidence for or against behavioral use. We report the pilot only to delimit the
 156 evidence and do not include it in the main contribution.

157 5 Limitations, responsible use, and conclusion

158 This is a one-model, one-benchmark case study. We did not run the requested second-family or
 159 second-scale replication, so the title and claims name Qwen2.5-Math-1.5B-Instruct. The transi-
 160 tion probe, temporal randomization, prefix and metadata controls, trace-weighting analysis, and
 161 subgroup analyses are post-hoc. No dated public preregistration exists; the repository records the
 162 frozen split seed, target, model-selection grid, and original bootstrap count, but cannot retroactively
 163 establish preregistration. Corrected/error twins have not received independent human verification,
 164 so no counterfactual-patching result is reported. A stronger next experiment would verify such twins,
 165 hold problem and boundary fixed, reserve an untouched onset test set, and validate the behavioral
 166 readout with a known effective intervention before testing a probe direction.

167 The detector is unsuitable for grading, certification, or unattended feedback. Its 38.8% false-alarm
 168 rate on correct traces could penalize valid work, and its late detections could allow an invalid deriva-
 169 tion to propagate. The source, generator, outcome, and length sensitivity makes deployment shift

170 especially risky. Positive uses are limited to research diagnostics with human or formal verification
171 of consequential mathematical claims.

172 The central empirical finding is a gap, not an internal “knowledge” claim. In Qwen2.5-Math-1.5B-
173 Instruct, invalid-so-far prefixes are linearly decodable and their scores change near annotated errors.
174 Yet nuisance metadata nearly matches pooled AUROC, exact onset localization is low, calibration
175 varies across trace length and source, cross-model generalization is unknown, and no behavioral
176 assay supports a causal conclusion. Reporting all of these together gives a more auditable account
177 of what hidden-state verifier scores do and do not establish.

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240 A Additional data accounting

Table 4: Attempted / retained traces / exclusions by source and split.

Split	GSM8K	MATH	OlympiadBench	Omni-MATH
Train	236/236/0	597/592/5	602/594/8	606/592/14
Validation	80/80/0	198/198/0	204/202/2	199/197/2
Test	84/84/0	205/204/1	194/188/6	195/193/2

241 The retained boundary counts by split and source are, respectively: train 1,229 / 3,718 / 5,129 /
 242 4,868; validation 414 / 1,303 / 1,708 / 1,555; and test 439 / 1,387 / 1,661 / 1,498 in the source
 243 order of the table. The original attempted total was 25,697 boundaries; 24,909 remain after the 40
 244 complete-trace exclusions.

245 B Descriptive subgroup audit

Table 5: Trace-macro outcomes by ProcessBench source (95% whole-trace bootstrap intervals).

Source	Complete accuracy	Exact error	Correct rejection
GSM8K	.512 [.405,.619]	.238 [.116,.375]	.786 [.658,.902]
MATH	.500 [.426,.569]	.347 [.264,.435]	.723 [.620,.818]
OlympiadBench	.298 [.234,.362]	.207 [.138,.282]	.463 [.340,.585]
Omni-MATH	.358 [.290,.425]	.324 [.253,.401]	.467 [.319,.614]

246 Among traces with an incorrect final answer, complete accuracy is 0.292 ([0.245, 0.342]). Among
 247 traces with a correct final answer it is 0.506 ([0.451, 0.554]); this group mixes valid traces and traces
 248 whose reasoning contains an annotated error. Across generators with at least 20 held-out traces,
 249 complete accuracy ranges from 0.316 to 0.621. These wide, overlapping estimates are descriptive
 250 evidence of heterogeneity rather than generator rankings.

251 **C Reproduction record**

252 The dataset SHA-256 is 447f0a4b35c5747a9f9a3dab1e70d43
253 f71efd501497b8cec668b71337099784a. All stochastic analysis uses seed 42. Exact re-
254 solved YAML files and artifact identity files are included with the anonymous supplement. The
255 principal commands are:

```
256 uv sync --extra dev
257 uv run math-error-tracing \
258   --config configs/experiment1.yaml validate-config
259 uv run math-error-tracing \
260   --config configs/experiment1.yaml run-all
261 uv run math-error-cpu-followup \
262   --config configs/experiment2.yaml
263 uv run math-error-extended-followup \
264   --config configs/experiment3.yaml \
265   fit-transition-probe
266 uv run pytest
267 uv run ruff check src tests
```

268 Activation extraction and the original intervention used one A100 GPU; probe fitting and all audit
269 analyses are CPU-capable. Extraction is sharded and resumable. The code rejects mismatched
270 dataset, model, dtype, and context-length identities and stores all selected thresholds and predictions.

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